

Special Weed Problems

[BARNYARDGRASS\(/PMG/WEEDS/barnyardgrass.html\)](#) AND VOLUNTEER CEREALS

These weeds are especially a problem in fields not preirrigated. Barnyardgrass is one of the most common weeds in California corn. High populations of this weed can occur in fields in continuous corn or those irrigated up. A high population of barnyardgrass may attract an infestation of armyworms, which moves from the weed to feed on corn leaves.

Preplant incorporated herbicide options for the control of barnyardgrass include s-metolachlor (Dual Magnum), alachlor (Micro-Tech), and EPTC (Eradicane).

In certain areas and soil types, pendimethalin (Prowl) can be applied to prevent any further germination of barnyardgrass seeds following the last cultivation for weeds. Thoroughly and uniformly incorporate pendimethalin into the soil with a sweep type or rolling cultivator set to incorporate in the top 1 inch of soil, and irrigate beds to thorough wetness.

The most effective postemergent treatment for control of barnyardgrass is to apply nicosulfuron (Accent) for control up to 3 inches in height or foramsulfuron (Option) for control up to 4 inches. A competitive corn crop and cultivation will improve control.

In a Roundup Ready tolerant crop, glyphosate can be applied over-the-top until the V8 stage for grass control. Barnyardgrass can be difficult to control, however, if it is drought stressed.

[NUTSEDGE\(/PMG/WEEDS/nutsedge.html\)](#)

Nutsedge is very competitive with corn in the early stages of growth. Running a sweep 4 inches below the top of the bed or power tilling the beds with L-shaped knives 3 to 4 inches deep before planting is effective in inhibiting nutsedge that has already sprouted. It is important to cultivate a second time for nutsedge after the crop has emerged. In the second cultivation, throw soil to the corn plants to help suppress nutsedge growth and allow corn growth to shade the furrow.

S-metolachlor (Dual Magnum) and alachlor (Micro-Tech) herbicides can give effective preemergent yellow nutsedge control. EPTC (Eradicane) controls both yellow and purple nutsedge. Use the highest label rates for your soil type for the most effective control.

Postemergent applications with halosulfuron (Sandea) or halosulfuron plus dicamba (Yukon) give good control of both purple and yellow nutsedge. In a Roundup Ready tolerant crop, glyphosate combined with close cultivation with sweep type cultivators can also give effective control of nutsedge.

[JOHNSONGRASS\(/PMG/WEEDS/johnsongrass.html\)](#)

Johnsongrass is a major problem in certain areas of California, especially where soils are difficult to dry down. Prevention is the key to johnsongrass control. Be especially diligent to prevent seedlings from becoming established. Once this weed is established on a farm, it is very difficult to eradicate. Johnsongrass seeds can get into irrigation water, manure, or lagoon water, and are easily spread throughout the farm. Ways to prevent its introduction are to ensure that feed is not contaminated with johnsongrass and to only use manure known to be well composted and free of weed seeds.

If a johnsongrass infestation does occur, there are several options to choose from in the management of this weed. In certain areas of the state, summer fallowing with a tillage operation every 3 to 4 weeks will give fair to good control by reducing carbohydrate reserves in rhizomes. Dry fallow is not effective in areas with a high water table where moisture in the soil is high enough throughout the summer to allow this weed to grow.

Glyphosate (Roundup) used in fallow periods, between crops, or when Roundup Ready (RR) varieties are grown can give effective control of rhizome johnsongrass during the growing season providing there is good soil moisture and active growth. Once glyphosate has been applied, allow about 7 to 10 days before disking to allow it complete translocation throughout the plant.

Preplant herbicide EPTC (Eradicane) can give rhizome suppression. Use the highest rate labeled for the soil type. EPTC, s-metolachlor (Dual Magnum), and alachlor (Micro-Tech) herbicides are effective in controlling johnsongrass seedlings before they emerge.

Nicosulfuron (Accent) as a postemergent banded treatment is very effective at controlling johnsongrass up to 1 foot tall and stopping the growth of johnsongrass up to 3 feet. Best corn yields are obtained by early (3- to 5-leaf) applications of nicosulfuron, which reduce early johnsongrass competition. If the first application time is missed, weed competition can be reduced by applying nicosulfuron with drop nozzles when the corn is 2- to 3-feet tall to control secondary flushes of seedlings in the row. As an additional measure, cultivate with sweep-type cultivators traveling at high speeds (5 to 6 mph) to throw 3 to 4 inches of soil to the base of the plant. The herbicide may also be applied as a band application in the corn row; johnsongrass growing in the areas between the rows can be destroyed by cultivation. Some growers may prefer to reduce weed competition by cultivating the furrows and applying this material with drop nozzles when the corn is 2- to 3-feet tall. This effectively controls secondary flushes of weed seedlings in the row. Cultivating the furrows after application and avoiding the use of nicosulfuron year after year in the same field will help prevent the development of resistance in johnsongrass to this herbicide.

Foramsulfuron (Option) is labeled for postemergent control of both seedling and rhizome johnsongrass up to 16 inches tall.

Cultivating the furrows after application and avoiding the use of nicosulfuron year after year in the same field will help prevent the development of resistance in johnsongrass to this material.

Rotating to broadleaf crops where clethodim (Prism), sethoxydim (Poast) and fluazifop-p-butyl (Fusilade) can be used is helpful in controlling johnsongrass between corn crops as can rotating to a Roundup Ready crop, such as alfalfa or cotton, where Roundup can be used for control.

BROADLEAVES

Some of the common broadleaves in corn include [redroot pigweed\(/PMG/WEEDS/redroot_pigweed.html\)](/PMG/WEEDS/redroot_pigweed.html), [common lambsquarters\(/PMG/WEEDS/lambsquarters.html\)](/PMG/WEEDS/lambsquarters.html), Palmer amaranth, annual morningglory, [common sunflower\(/PMG/WEEDS/sunflowers.html\)](/PMG/WEEDS/sunflowers.html), [common purslane\(/PMG/WEEDS/purslane.html\)](/PMG/WEEDS/purslane.html), horse purslane, [annual morningglory\(/PMG/WEEDS/morningglories.html\)](/PMG/WEEDS/morningglories.html), [cocklebur\(/PMG/WEEDS/common_cocklebur.html\)](/PMG/WEEDS/common_cocklebur.html), [hairy nightshade\(/PMG/WEEDS/hairy_nightshade.html\)](/PMG/WEEDS/hairy_nightshade.html), [black nightshade\(/PMG/WEEDS/black_nightshade.html\)](/PMG/WEEDS/black_nightshade.html), [velvetleaf\(/PMG/WEEDS/velvetleaf.html\)](/PMG/WEEDS/velvetleaf.html), and [field bindweed\(/PMG/WEEDS/field_bindweed.html\)](/PMG/WEEDS/field_bindweed.html).

Preplant herbicides alachlor (Micro-Tech) and S-metolachlor (Dual Magnum) are mainly grass herbicides but control some broadleaves such as nightshades and purslane. EPTC (Eradicane) controls the same weeds but also controls annual morningglory and lambsquarter.

Cultivation is important in controlling many small broadleaves. Broadleaves that emerge after the first irrigation when it's not possible to cultivate again because the corn is too tall can be controlled postemergent with 2,4-D. Dicamba (Banvel, Clarity) controls the same weeds and, in addition, is more effective on purslane, velvetleaf, and field bindweed. Dicamba and 2,4-D are low cost options but be careful because they can easily drift or volatilize onto sensitive crops and cause severe injury.

Bromoxynil (Buctril) and carfentrazone (Shark), both contact herbicides, are other postemergent option. For bromoxynil to be effective, weeds must be in the seedling stage or no larger than the size of a quarter. Carfentrazone, on the other hand, is effective on larger weeds (see label for details). These herbicides are a good choice to use as buffers near sensitive crops. Bromoxynil alone does not do an adequate job of controlling redroot pigweed.

Foramsulfuron (Option) is labeled for postemergent control of many broadleaves up to 2 to 3 inches tall.

Glyphosate (Roundup) for use on Roundup Ready (RR) varieties of corn is very effective in controlling many broadleaf weeds. It can also be tank mixed with selective herbicides; see label for RR variety information.

Metribuzin (Sencor) is a postemergent contact herbicide that is very effective at controlling broadleaf seedlings but does have some crop safety issues (see label).

Nicosulfuron (Accent) is very effective at controlling many small broadleaf seedlings after they germinate. It can be tank mixed with metribuzin, dicamba, or bromoxynil to broaden the weed control spectrum. Nicosulfuron is best used in conjunction with other methods of control including preplant incorporated herbicides and cultivation of furrows.

RESISTANT WEED MANAGEMENT ISSUES

An integrated approach to weed management is important in maintaining herbicide effectiveness. In some areas, use of EPTC (Eradicane) in the same fields year after year can lead to a decrease in the level of performance of this herbicide. The decrease is caused by a buildup of microbes in the soil that rapidly degrade the herbicide. To prevent this, use another preplant herbicide with different chemistry, such as alachlor or s-metolachlor. Also, avoid using nicosulfuron (Accent) or foramsulfuron (Option) in the same field for more than 2 consecutive years. Sulfonylurea weed resistance has developed rapidly because its mode of action is at a single site. Tank mixing with another herbicide, along with the use of cultivation in the furrows, will also reduce the risk of resistance developing to nicosulfuron. Use crop rotation or an herbicide of different chemistry if corn is grown several years in the same field.

In Roundup Ready crop systems in other states, weed shifts and weed resistance occurs. Weed shifts can occur when an herbicide program is used repeatedly, resulting in the survival of only weeds that are tolerant of the herbicide. Weed shifts also occur when the susceptible portion of a particular weed's population is controlled, but resistant biotypes of the weed survive and gradually produce a population that is resistant to the herbicide. Weed shifts are usually associated with reduced tillage systems and not rotating herbicides.

A major concern is the development of resistance to glyphosate (Roundup) by lambsquarter, amaranth, horseweed, and Italian ryegrass in California. Rotating glyphosate-resistant corn with another glyphosate-resistant crop such as cotton or alfalfa will increase this problem.

To help prevent the development of herbicide-resistant weeds and prevent weed shifts from occurring, it is important to incorporate tillage into your weed management practices as well as alternating herbicides that have a different chemical mode of action. Mode of action numbers are listed for the different herbicides used in corn in the [HERBICIDE TREATMENT TABLE\(/agriculture/corn/herbicide-treatment-table/\)](https://ucanr.edu/About/DirectorySearch/?facultyid=2495).



UC IPM Pest Management Guidelines: Corn
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